HireNeuron Verity

Evidence-First Autonomous AI Interviewer — Complete Final Solution

25 October 2026

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Evidence-First Autonomous AI Interviewer — Consolidated Final Solution
Prepared: 25 October 2026
Document type: Judge-facing concept + detailed internal implementation specification
Implementation status: Proposed architecture. Live performance, detection accuracy, cost, scale and provider integration must be established by testing.

Executive recommendation

Keep **HireNeuron Verity**, but simplify the implementation and strengthen its real-time reliability. The product is a recruiter-configurable platform that conducts a complete voice interview from a job description and candidate resume; supports adaptive follow-ups; independently observes two consented camera feeds; stores reviewable integrity evidence; and produces transcript-grounded assessments with recruiter override. The solution deliberately avoids calling observations definitive cheating evidence or automatically making hiring decisions.

Core principles:

1. **Deterministic interview control:** Software owns question state, duration, timeouts, retries and completion; the LLM handles question wording, follow-ups and assessment.
2. **Server-verified two-camera proctoring:** Both videos are sampled by a trusted backend rather than trusting client anomaly messages.
3. **Evidence rather than accusations:** Observable events, suspicious patterns and monitoring gaps have separate labels; only a human may judge concerns.
4. **Human hiring decisions:** Transcript-grounded AI scores inform a recruiter, not replace one.

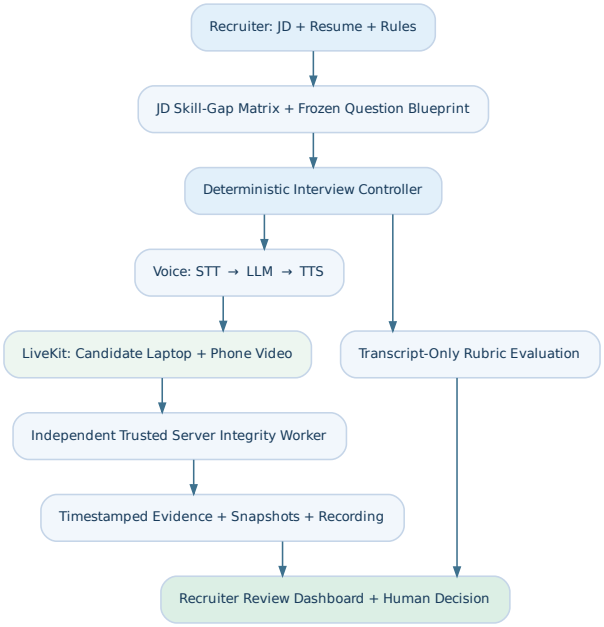
The official problem statement specifically prioritizes an *integrated prototype*, rather than merely a question-generation chatbot. The solution covers resume/JD analysis, configurable and default questions, real-time voice interaction, secondary phone camera, event logging, post-interview rubric evaluation, and recruiter reports.

Updated decisions on the external review

Review concern	Implemented architectural decision
Redundant primary-camera CV chain	**Accept:** remove the browser-hint plus random-audit trust chain from P0; sample both feeds directly on the server at approximately 1 FPS. Keep browser MediaPipe only for positioning UX.
LLM TTS may be slow	**Accept:** measure time-to-first-audio with Gemini Flash-Lite TTS; use a provider abstraction and a low-latency Deepgram fallback if required.
Unreliable conversational turn-taking	**Strongly accept:** voice activity detection, semantic end-of-turn inference, adjustable endpointing, and an explicit Done Answering control.
Raw-track synchronized video replay complexity	**Accept:** signed snapshots and one browser-playable event clip or composed recording first; full synchronized raw-track playback later.
STT, TTS, LLM and dual-camera CV in one worker	**Accept:** measure simultaneous load; run integrity in a separate process/container when necessary, without starving interview speech.
Spoken skip/repeat undefined	**Accept:** explicit phrase checks, confirmation for ambiguous skip, persistent question statuses, and visible manual controls.
Model identifiers and prices need recheck	**Accept:** version provider IDs and recheck pricing, quotas and availability at build/submission. Never claim an unmeasured fixed session cost.
Long, review-history-heavy submission	**Accept:** use the executive material as the judge-facing summary; retain the following sections as the detailed implementation contract.

Additional decision: Attempt a simple browser-compatible recording or event clip earlier than advanced evidence sealing, because recording review appears in the official requirements. Do not portray unimplemented replay as working.

End-to-end architecture



Recommended stack (subject to integration and latency tests):

Layer	Technology / scope
Candidate, phone and recruiter UI	Next.js, TypeScript, Tailwind CSS
API + interview state controller	Python FastAPI
Realtime media	LiveKit Cloud / WebRTC
Streaming transcription	Deepgram Nova-3
Interview questioning and normal scoring	Gemini 3.8 Flash
Optional difficult assessment	Gemini 4 Argon, asynchronous only
Question speech	Gemini 3.8 Flash-Lite TTS after benchmark; Deepgram Aura-2 / Flux fallback
Voice turn detection	VAD + LiveKit audio turn detector + adaptive endpointing
Integrity vision	Server-side face/person detection using MediaPipe or suitable ONNX runtime
Storage and session state	PostgreSQL, Redis, encrypted object storage
Evidence capture	Server snapshots, timestamped event manifests, optional LiveKit Egress recording
Runtime	Managed frontend, always-on agent, independent integrity process, asynchronous evaluation

What makes the prototype convincing

Intelligent rather than generic interviewing. Generate a frozen, recruiter-controlled rubric from the job description; use the candidate's resume only to decide what claim to probe. Ask one grounded, time-budgeted follow-up rather than creating endless extra questions.

Trust without false accusation. A short glance away is an observation. A sustained, server-confirmed extra person becomes high-priority *review*, not a fraud verdict. Failed camera/CV coverage is recorded as an *unverified interval*, never used to penalize a candidate.

Traceable scoring. Every score links to a real answer span; alleged missing concepts remain explicitly model-inferred and must be checked against transcript evidence. Accent, facial expression and proctor alerts are excluded from technical grading.

Recoverable workflows. An LLM outage does not destroy a frozen question blueprint; phone disconnection does not reset the interview; failed integrity inference marks missing coverage transparently.

Build priorities in order

Milestone	Non-negotiable output
1 — Reliable interview core	Recruiter input, secure document parsing, frozen questions, text state-machine testing, streaming voice, skip/repeat/timeout handling.
2 — Phone and independent feeds	One-use QR, separate consent, phone video-only credential, laptop and phone both visible and subscribed independently.
3 — Server integrity	~1 FPS each feed, face absence and multiple-person observation, repeated-event logic, frame snapshots and device loss telemetry.
4 — Recruiter review	Per-question transcript spans, skill matrix, rubric scores, evidence timeline and append-only override history.
5 — Reliability and playable media	Dual-feed load test, TTS/turn latency tuning, reconnect recovery, one real recording/clip and optional Phase B sealing.

The team should pass the first four milestones before investing in sophisticated object detection, identity checks, dual raw-track synchronized playback, lip-sync forensics, photorealistic avatars or distributed infrastructure.

Final assessment and constraints

Dimension	Plan assessment (subjective, not tested)
Official problem statement alignment	**9.5 / 10**
Architectural design	**9 / 10**
Prototype build feasibility	**8 / 10**

The principal risks are natural live turn-taking, performance with both camera decoders active, network resilience, and event-detector false positives. A 1 FPS prototype cannot catch every short-lived activity; camera coverage cannot reveal hidden off-screen help; tamper-evident server files do not prove the physical room is authentic. **Do not claim 100% cheating detection, fixed cost, perfect fairness or production scalability.**

Full Implementation Specification

The remainder is the consolidated engineering build contract: required behavior, schemas, security, evaluation, resilience, test gates, demo sequence, limitations and references. The same principles and components above are specified in implementable detail.

Detailed system requirements and implementation

2. Solution scope and acceptance

Required P0

- Recruiter enters title, JD, skills, experience, difficulty, duration and question categories; uploads PDF/DOCX resume; defines mandatory questions. Blank count generates exactly 10 primary questions.

- Parse JD and resume separately. The JD/recruiter requirements define competencies, weights and rubrics. Resume claims are untrusted context that can produce probing questions but cannot modify grading policy. Show Match / Claim / Gap matrix.
- Freeze the primary question blueprint before starting; honor recruiter question text verbatim. Default 10-question mixture: 4 technical, 2 resume/project, 2 scenarios, 1 behavioural, 1 JD gap (configurable).
- Conduct an actual two-way voice interview with streaming transcription, question speech, no premature interruptions, explicit answer timers, optional bounded follow-up, deterministic skip/repeat/rephrase commands, and restore/reconnect.
- Pair smartphone through a one-use QR code with separate consent. Laptop camera + microphone remain active; phone publishes **video only** and cannot subscribe. Both videos are displayed and separately received on server.
- Sample both low-resolution feeds on the server around 1 FPS each, subject to measured device capacity. Primary: face present/missing, face/person counts, coarse head-pose indicators. Secondary: additional people. Camera/mic/phone interruptions produce coverage-status events. Proctor status is observational, never a cheating verdict.
- Show recruiter report with exact answer excerpts, question status, rubric score and strengths/gaps, event timestamps, server snapshots, visibility/coverage limitations, and append-only manual score or event overrides.
- Keep original evidence snapshots, signed event records, consent records, and runtime audit logs. Provide at least one authorized playable session recording or event clip if possible; keep synchronized separate-track replay out of P0.

Prioritized optional P1

- Composed-room MP4 recording with two views, post-session per-event MP4 clips and Phase B final recording hash/seal; degraded phone HTTPS snapshots; audio anomaly review; additional-object detection; stronger automatic camera framing checks.

Roadmap P2

- Independent raw-track multi-angle synchronized playback, client hint verification for production cost tuning, sophisticated device/phone/object CV, screen share, biometric identity continuity (where consent and law permit), extensive diarization and liveness, distributed scaling and compliance attestations.

3. Component architecture

1. **Web:** Next.js + TypeScript + Tailwind: recruiter, candidate, and mobile companion UI.
2. **Control API:** FastAPI; session orchestration, resume/JD ingestion, permissioned endpoints, role access, rubric and question contracts.
3. **Data:** PostgreSQL authoritative store; Redis for expiring QR challenges and short-lived checkpoints; private encrypted object storage for evidence and recordings.
4. **Realtime transport:** LiveKit Cloud WebRTC. Distinct room participants/tracks for candidate camera/mic, phone camera, AI speech, with explicit roles. Do **not** rely on agent `video_input=True` for two camera streams: it follows the most recently published video.
5. **Interview agent:** Always-running LiveKit Agent, with Deepgram Nova-3 English STT + vocabulary hints, Gemini 3.8 Flash for questions/follow-ups/scoring, and a selected streaming TTS. Pre-generate primary questions. Gemini 4 Argon is optional for post-session adjudication, not a hot-path dependency.
6. **Independent integrity process/worker:** Explicitly subscribes to both role-bound video tracks, resizes, samples, runs detectors, maintains small timestamped ring buffers, sends evidence events. Start separate from voice in production-like P0 to avoid media processing starving turn-taking; it can share a VM but should remain a separate process.
7. **Asynchronous evaluation process:** scores transcript answer spans against frozen rubric only after/away from real-time conversation. A failure does not interrupt the session.

Data flow: Recruiter JD/resume/config → frozen skill matrix and question blueprint → deterministic state machine → LiveKit voice interaction and parallel two-camera verification → versioned transcript & evidence → isolated evaluation → recruiter report and human decision.

4. Interview intelligence, answer control and timing

- **States:** CREATED → BLUEPRINT_FROZEN → CONSENT → DEVICE_CHECK → PHONE_PAIRING (optional where policy allows) → READY → ASKING → LISTENING → TURN_DECISION → NEXT_SLOT / FOLLOW_UP → COMPLETED → EVALUATING → REVIEW_READY. Recover state using server-owned session and slot IDs.
- **Question statuses:** PENDING, ASKING, ANSWERING, COMPLETED, SKIPPED, UNANSWERED, TIMED_OUT. Follow-ups are children of a primary slot, never additional primary slots.
- **Default duration:** 25 min; recruiter may configure another duration. The total includes onboarding? Define timer as starting on READY/first question; configuration should explicitly display time budget. Have a closing reserve and per-category answer caps. Hard cutoff rules belong in server state machine, not LLM.
- **Follow-up:** max one per primary question, only when answer needs probing and time reserve allows; if provider is down, continue to frozen next primary question.
- **Speech:** start with Deepgram Nova-3 streaming, `interim_results` for captions; combine finalized chunks and never equate an interim fragment with a completed answer. Use VAD + LiveKit audio turn detector and dynamic endpointing; use roughly 0.3–2.5s as initial *tuning range*, not guaranteed latency or fixed universal optimal settings. For candidates with mid-sentence pauses, wait longer unless there is strong end-of-turn evidence. Give a visible 'Done answering' button as explicit user override. Do not repeatedly send finished transcripts to the LLM as new turns.
- **Barge-in:** candidate may politely interrupt AI question playback to ask to repeat, but incidental noises must not interrupt TTS. If interrupted, cancel playback cleanly and maintain slot state.
- **Skip/repeat:** normalize final STT text. Recognize short high-confidence explicit command phrases, e.g. 'skip this question', 'please repeat', 'rephrase'. A bare or ambiguous 'skip' prompts 'Do you want to skip this question?' and requires a yes/no confirmation before writing SKIPPED. Button controls always available. Repeating does not consume a question or alter answer status.
- **Silence:** first gentle reminder, then ask repeat/rephrase/skip; if answer deadline arrives with no content, label UNANSWERED; with incomplete/expired answer, label TIMED_OUT and preserve transcribed material. Do not silently give zero without recruiter-visible explanation.
- **STT failure:** temporary voice retries and transparent transcript coverage marker, not fabricated text; use visible text fallback or repeat request if candidate content was not captured.
- **TTS test gate:** benchmark time from final candidate word to first interviewer audio and P95 latency over representative utterances. Start with `gemini-3.8-flash-lite-tts` only if latency is acceptable (target ~1.5s for common short questions); pre-cache introductory and fixed primary question speech. Have `deepgram/aura-2` or `deepgram/flux-tts` ready as a provider-swappable fallback. Compare under actual region/network conditions, not provider marketing numbers.

5. Secure phone pairing

- Generate cryptographically random 256-bit opaque QR challenge, store only hashed challenge plus interview/tenant/device/expiry data in Redis for ~180 seconds.
- Show mobile page with secondary camera and analysis consent *before* publication; validate one-use nonce atomically; issue short-lived, narrowly scoped LiveKit publish-video-only credential. Disallow publishing microphone, joining other rooms or subscribing to tracks. Do not put a room-wide JWT into QR.
- Server validates interview owner, active session and role; rate-limits redemption; expires unused nonces; logs policy version, timestamps and roles. Provide reconnect token renewal safely after initial redemption.
- Guide candidate to position phone so upper body, hands, desk and nearby workspace are visible, where physically possible. Never claim the coverage is complete.

6. Integrity detector and evidence trust

Trusted sampling

- On the trusted integrity service, create two independent video readers keyed by **server-validated** PRIMARY_CAMERA and SECONDARY_CAMERA roles. Never treat browser-provided arbitrary identity labels as trusted without session authorization.
- Infer on one frame per second per feed as baseline, at a low but verified useful resolution; adapt down on overload and log coverage gaps. Keep an immutable server receive timestamp and a small frame ring buffer for surrounding event snapshots. Do not silently downgrade coverage.
- PRIMARY: face presence/absence; person/face count; coarse head-pose category. SECONDARY: person count/sustained new person. Choose licensed, lightweight ONNX/MediaPipe implementations and benchmark against actual demo devices. 'Gaze' is only a rough attention indicator; webcam head direction cannot establish reading an outside screen.

Taxonomy

- `OBSERVED_EVENT`: objective event only, e.g. brief face absence, head turn, browser-hidden client hint, brief disconnect.
- `NEEDS_REVIEW`: sustained face absence, calibrated repeated attention changes, prolonged media interruption.
- `NEEDS_REVIEW_HIGH`: repeated, server-corroborated additional person visible in primary or secondary scene. Even this is **not proof of cheating**.
- `UNVERIFIED`: missing frames, degraded/snapshot camera mode, worker failure, insufficient confidence.

Do not combine all small warnings into an opaque 'cheating percentage'. Use explicit evidence, intervals and confidence/coverage descriptions. Thresholds must be set with actual labeled benign versus simulated-integrity scenarios, not invented accuracy rates.

Audio handling

- The primary mic serves STT. P0 records microphone availability and timestamps of unusual interruptions/crosstalk where reliably observable; it does **not** infer a 'second person' from ordinary VAD alone.
- Any later voice anomaly model must be tested on recorded consented samples, must suppress the AI's own TTS audio plus measured echo tail, and may at most produce `OBSERVED_EVENT` without independent evidence. Phone mic stays disabled.

Evidence records

- A detection creates a server event with ID, session ID, UTC timestamp + monotonic per-session offset, camera role, algorithm/model version, observed type, severity, detection/coverage confidence, frame object path and SHA-256, and previous event record hash.
- **Phase A (P0):** persist encrypted server-captured snapshots (possibly multiple over a time window), commit the event and hash-chain or HMAC the manifest using a server-only secret. Do not sign a file before upload and assume it exists; store and verify object checksum first, then seal.
- **Phase B (P1):** create one browser-compatible *composed* MP4 or encoded event clip with both views through LiveKit Egress or permitted post-session processing. After finalize, link object ID, actual file SHA-256, recording start time and event-relative seek offset to signed event in a second signed manifest. Do not promise synchronized raw-track playback unless measured and implemented.
- Recording failure does not invalidate snapshots: label RECORDING_INCOMPLETE; evidence trust is **tamper-evident from server receipt**, not unforgeable evidence of the physical room.

7. Evaluation and recruiter dashboard

- Score the **frozen job rubric** against **only** user answer transcript spans. No face features, gender, accent, speaking rate, background, head pose, or proctoring event severity is passed into the answer scorer. In interview settings, 'communication' means understandable, logical explanation, not grammatical polish.
- Each slot stores 1–5 technical/role-relevant scores as configured and an explanation that references exact `transcript_span_id` values. Model outputs must pass JSON-schema and span-existence checks. Any inferred missing concept is clearly marked as model-inferred absence and checked against available spans to avoid contradictory claims.
- Missing/unanswered or poor transcription is labeled insufficient evidence, not automatic incompetence. Allow review or re-interview recommendations.
- Report shows candidate/job data, question list and statuses, follow-ups, annotated answer excerpts, skill gap matrix, per-question and summary rubric assessment, evaluated model version, and independent proctor timeline with evidence thumbnails and source timestamps.
- Recruiter can dismiss or confirm review concerns and change AI scores, recording the reviewer, reason and original score in append-only audit history. Show a neutral outcome scale such as Strong / Moderate / Insufficient Evidence / Needs Further Assessment; no auto-hire/auto-reject decision.

8. Resilience and deployment

- Web: managed Next.js host. Control: FastAPI on Cloud Run or VM. Database: managed PostgreSQL; Redis checkpoint/token TTL. Media: LiveKit Cloud. Voice: LiveKit managed agent or always-on worker. Integrity: distinct always-on process/container; evaluation: asynchronous worker. Do not use an ordinary Cloud Run request service for a permanently connected background media subscriber without lifecycle design.
- Test with both feeds, STT, LLM and TTS at once, recording CPU, memory, FPS, event queue delay and P50/P95 voice latency. Document **one** verified concurrency capacity number rather than guessing 200 sessions per server.
- Handle reconnect without losing the question; no transcript is invented for coverage gaps. On primary mic loss, pause listening; on phone loss, emit interval event and follow policy; on detector failure, mark UNVERIFIED; on LLM failure, ask frozen questions; on TTS failure, display text while retrying. If candidate reloads, restore blueprint/slot progress.
- Respect participant roles, access controls, encryption, storage retention and deletion policies. Capture explicit separate laptop/phone consent before recording/analysis; provide authorized access only. No blanket legal-compliance claim.

9. Measurements and release gates

Pass before demo: 1. Empty count -> exactly 10 primary questions; custom questions unchanged; poisoned resume text cannot mutate grading rubrics. 2. Five continuous spoken Q/A turns with no unwanted early AI interruptions, correct transcript-slot attachment and bounded contextual follow-up. 3. 'Skip' ambiguous -> confirmation; confirmed skip -> SKIPPED; repeat -> same slot; silent response -> UNANSWERED; partial expired response -> TIMED_OUT. 4. Phone joins and primary never disappears; both live thumbnails update and server frame counts for each role rise. 5. Additional person on phone feed persists long enough -> `NEEDS_REVIEW_HIGH` with real multi-frame server images; brief head turn does not accuse cheating; CV outage -> `UNVERIFIED`. 6. Mic/camera/phone loss logs event with start and end; reconnect returns to current slot; already-scored questions do not duplicate. 7. Report cites exact actual transcript spans; missing claims validated; reviewer override preserves original values; evidence retrieval checks access. 8. Video/media stress test with two feeds and voice is stable under venue/hotspot conditions. If extra video load impairs conversation, lower inference FPS or dedicate integrity compute and openly show coverage reduction. 9. If recording is shown, verify real playable output and seek offsets; never mock a synchronized clip as if implemented. 10. Benchmark actual STT error rate on technical terms/Indian English and sample transcription, event false-alarm rate on consented test clips, TTS first-audio latency and overall perceived turn latency. Treat desired thresholds as targets until measured.

10. Demo plan (approximately five minutes)

- **0:00–0:40 Configuration:** upload sample JD/resume; show gap matrix, pre-generated 10 questions and a locked recruiter-defined question.
- **0:40–1:20 Pairing:** show laptop consent; scan real phone QR; grant phone consent; show both continuously updating feeds and independent role-bound server receipt.
- **1:20–2:35 Intelligent interview:** candidate answers a project-specific question; AI asks one contextual follow-up; demonstrate repeat or skip confirmation and transcript state.
- **2:35–3:40 Evidence:** benign brief look-away -> observation, not accusation; simulated additional person enters phone field of view for a calibrated duration -> review-high with multiple real server snapshots; optionally show phone disconnect and coverage gap.
- **3:40–4:50 Report:** open a **clearly identified, previously recorded actual test session** for question scores, quoted answer spans, event snapshots/clip if working, and recruiter override. Finish with principle that the recruiter makes the decision.

11. Realistic remaining limits

- No camera-only design reliably detects hidden hardware, invisible remote help, external second monitors or sophisticated virtual camera substitution.
- At 1 FPS, short transient events may be missed and reliable eye-gaze is not possible; coverage must be reported. Low-frame-rate sampling is chosen for the prototype's responsiveness, not claimed as high-recall surveillance.
- Voice diarization and accent fairness require actual multi-speaker and regional-English test data; labels and calibrated confidence need empirical validation.
- A server-signed media record proves integrity **after receipt**, not that the physical scene was authentic.
- Vendor availability, model pricing, rate limits and local legal requirements must be checked at deployment/submission time.

12. Selected verified technology references (as of 25 Oct 2026)

- Google Gemini model names: https://ai.google.dev/gemini-api/docs/models
- Google pricing: https://ai.google.dev/gemini-api/docs/pricing
- LiveKit audio turn detection and tuning: https://docs.livekit.io/agents/logic/turns/turn-detector/
- LiveKit deepgram TTS integrations: https://docs.livekit.io/agents/models/tts/deepgram/
- LiveKit two-video automatic input limitation: https://docs.livekit.io/agents/multimodality/vision/video/
- LiveKit Egress recording: https://docs.livekit.io/transport/media/ingress-egress/egress/
- Deepgram endpointing: https://developers.deepgram.com/docs/understand-endpointing-interim-results/

Submission posture: Short judge-facing architecture + one polished diagram, known limitations, demo script. Keep this blueprint as the engineering contract; never submit untested numbers or claim future features already work.